CHAPTER 5
EPIC 2: CORE FUNCTIONALITIES DEVELOPMENT

5.1 Introduction

This chapter explains the implementation of the major functional modules of EduGenie. The application was developed using Python, FastAPI, Google Gemini, and LaMini-Flan-T5. Each module performs a specific educational task while sharing a common backend architecture.

5.2 Question Answering Module

This module accepts academic questions from users and forwards them to Google Gemini through the FastAPI backend. The AI model generates context-aware answers that are displayed on the web interface. The module reduces search time and improves learning efficiency.

5.3 Explanation Module

The Explanation Module simplifies complex concepts into easy-to-understand language using LaMini-Flan-T5. It is intended for beginners and students who need conceptual clarity.

5.4 Quiz Generation Module

This module generates topic-based multiple-choice questions automatically. It supports self-assessment, revision, and exam preparation by creating interactive quizzes from user-specified topics.

5.5 Summary Module

The Summary Module converts lengthy educational content into concise notes. It extracts key ideas and presents them in a readable format, helping students revise quickly.

5.6 Learning Path Recommendation Module

This module generates personalized learning roadmaps for selected subjects. It recommends an ordered sequence of topics from beginner to advanced levels, allowing students to plan their studies effectively.

5.7 Backend Integration

All modules communicate through the FastAPI backend. Requests from the HTML frontend are validated, processed by the appropriate AI model, and returned as structured responses. Error handling and modular routing improve reliability.

5.8 Implementation Workflow

The workflow begins with user input through the web interface. FastAPI routes the request to the required AI module. The AI generates the response, which is formatted and displayed to the user. During the final report, insert your website screenshots after this section to demonstrate the working of each module.

5.9 Advantages

The modular implementation improves code reuse, simplifies debugging, supports future enhancements, and delivers fast educational responses through AI integration.

5.10 Chapter Summary

This chapter described the implementation of all core functionalities of EduGenie. The integration of FastAPI with Google Gemini and LaMini-Flan-T5 enables an efficient, scalable, and user-friendly AI-powered learning assistant.